ARMED CONFLICT AND FOOD INSECURITY: IV IDENTIFICATION IN A DYNAMIC BINARY PANEL*

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Abstract

Armed conflict raises the probability of acute food insecurity by 13 percentage points contemporaneously in a newly constructed district-by-country-by-quarter panel covering 1,162 districts across 21 developing and fragile countries from 2010 to 2024. The long-run multiplier implies a 22-percentage-point increase following a permanent conflict shock. Identification rests on a leave-one-out country-level conflict wave that isolates within-district variation for districts on the margin of country-level escalation (Imbens & Angrist, 1994). Because the food-crisis outcome is binary and conflict exposure is endogenous within a dynamic panel, we implement both the parametric IV-Probit of Rivers & Vuong (1988) and the semiparametric special regressor of Lewbel (2000), grounding both estimators in a formal latent-index data-generating process that resolves the initial-conditions problem for the binary outcome (Heckman, 1981). Convergence of the two IV estimators is consistent with the bivariate normality assumption underlying IV-Probit, subject to the special regressor's own conditional exogeneity. Mechanism analysis identifies reduced agricultural production as the dominant transmission pathway. Food prices counterintuitively *fall* in conflict-affected districts, consistent with demand collapse and humanitarian food aid inflows dominating supply-chain disruptions.

JEL Classification: C23, C26, D74, I31, O12

Keywords: food insecurity, armed conflict, binary endogenous regressor, special regressor, IV-Probit, dynamic panel, district-level panel, conflict waves

1 Introduction

Identifying the causal effect of armed conflict on acute food insecurity is an important empirical question, but it poses well-known identification challenges. Both variables are likely jointly determined by common unobservables and by each other: food insecurity can fuel grievances and resource competition that precipitate conflict (Hendrix & Brinkman, 2013), while conflict destroys agricultural capital, disrupts supply chains, and displaces populations in ways that drive food crises (Blattman & Miguel, 2010; Shemyakina, 2022). Standard panel regression therefore conflates the effect of conflict on food insecurity with the effect of food insecurity on conflict. Compounding this simultaneity challenge, the outcome of interest (whether a district enters acute food crisis) is binary, making standard linear IV estimators inconsistent and requiring either parametric distributional assumptions (IV-Probit) or semiparametric alternatives (Lewbel, 2000). This paper proposes an integrated approach to these challenges and implements it on a new high-frequency district-level panel.

Acute food insecurity has become a central concern in global policy debates, sitting at the intersection of the United Nations Sustainable Development Goals on poverty elimination, hunger, and population health (SDGs 1, 2, and 3). Despite sustained international attention, the number of acutely food-insecure people has risen sharply in recent years, especially in fragile and conflict-affected settings. In 2024, more than 343 million people faced acute food insecurity across 74 countries—about 10 million more than in 2023 and nearly 200 million more than before the pandemic (FAO et al., 2024). Roughly 44.4 million people were in emergency conditions or worse, including approximately 1.9 million on the verge of famine. In the same year, about 65 percent of acutely food-insecure people lived in fragile and conflict-affected situations, and all countries classified by the World Bank as fragile and conflict-affected experienced food security ratings of "in crisis" or worse.

Figure 1 illustrates the strong time-series co-movement between conflict events and food

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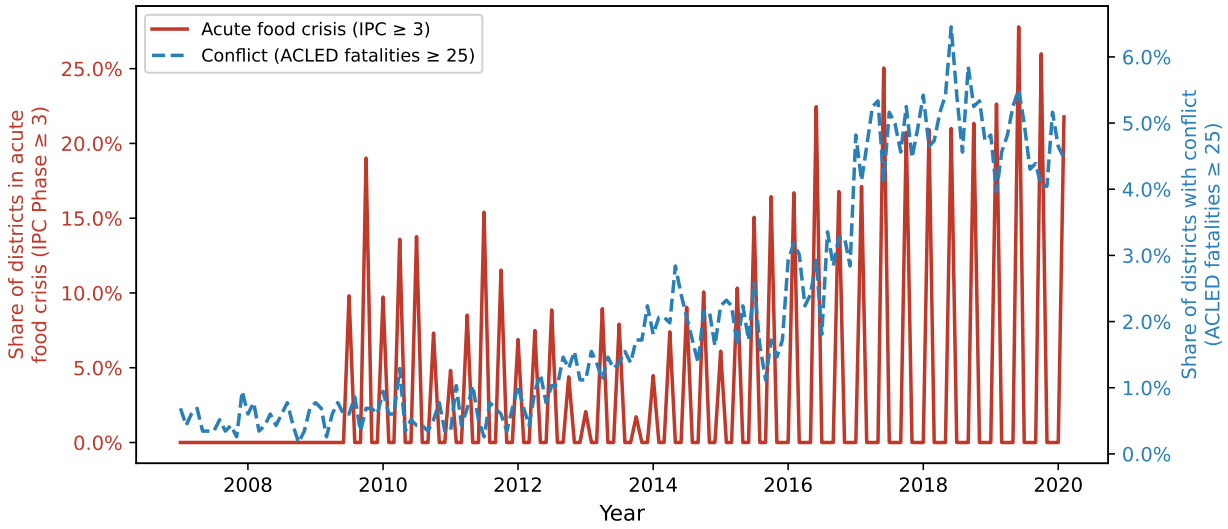


Figure 1: Conflict events and prevalence of food insecurity in fragile and conflict-affected states (FCS). The figure plots the annual count of ACLED conflict events and the share of the population facing acute food insecurity (IPC Phase ≥ 3) over time.
Source: ACLED and FEWS NET.

insecurity prevalence in fragile and conflict-affected states. While this correlation does not establish causality (and is indeed consistent with both causal directions) it motivates the need for empirical frameworks that can disentangle the direction, magnitude, and persistence of the conflict-to-food-insecurity channel.

A central empirical challenge is measurement. Both conflict and food insecurity are inherently local and evolve rapidly over time, yet much of the existing cross-country evidence relies on coarse spatial units (national or sub-national aggregates) or low-frequency outcomes that obscure the timing, geography, and persistence of shocks (Martin-Shields & Stojetz, 2019; Shemyakina, 2022). A complementary micro-level literature uses household and individual data with difference-in-differences or IV strategies in single-country settings (Verwimp et al., 2019), providing important causal evidence but with limited external validity. This paper bridges these two strands by constructing a panel that is simultaneously high-frequency (quarterly), fine-grained (district-level), and multi-country.

To answer this question at a scale that existing studies cannot, we construct a district-by-country-by-quarter panel linking geocoded conflict events from ACLED with IPC-based food

insecurity measures from FEWS NET, covering 1,162 districts across 21 developing and fragile countries from 2010 to 2024. This dataset enables a dynamic analysis of the conflict–food insecurity relationship (including persistence, lag structures, and transmission channels) at a spatial and temporal resolution unavailable in earlier cross-country work. Armed conflict significantly increases the probability of acute food insecurity both contemporaneously and persistently: in the preferred IV-Probit specification, the contemporaneous causal effect is approximately 13 percentage points, and the long-run multiplier implies a 22-percentage-point increase following a permanent conflict shock.

The combination of a binary outcome, an endogenous regressor, and a dynamic panel creates a non-standard estimation problem. We address it by first formalizing a latent-index data-generating process (DGP) from which all estimating equations follow as direct consequences, ensuring that identification assumptions are stated relative to structural model primitives rather than imposed post hoc. The DGP structure also clarifies the *initial conditions problem* (Heckman, 1981): the district fixed effect and the initial food-crisis observation are correlated by construction, invalidating standard within-MLE for binary outcomes (Neyman & Scott, 1948). Following Wooldridge (2005), we model this correlation explicitly and test for its empirical importance. The binary endogenous regressor is addressed by the parametric IV-Probit of Rivers & Vuong (1988), which assumes bivariate normality of the latent errors, cross-validated against the semiparametric special regressor of Lewbel (2000) (simplified by Dong & Lewbel 2015), which imposes no distributional assumption; their convergence serves as an informal test of the normality restriction. The leave-one-out conflict wave instrument identifies the local average treatment effect (LATE) for districts on the margin of country-level conflict escalation (Imbens & Angrist, 1994). Convergence of IV-Probit and special-regressor estimates across specifications is consistent with the bivariate normality assumption per Proposition 3; as that proposition states, convergence is necessary but not sufficient for distributional correctness.

Related econometrics literature. Endogenous binary regressors have been studied since at least Rivers & Vuong (1988), whose parametric IV-Probit remains the canonical solution, and Lewbel (2000), who showed that a large-support special regressor renders the binary outcome tractable without distributional assumptions (see Maddala, 1983, for a survey). We implement both and exploit their convergence as an informal distributional test. The initial-conditions problem in dynamic binary panels is a distinct challenge (Heckman, 1981): treating district fixed effects as unrestricted incidental parameters yields inconsistent MLE (Neyman & Scott, 1948), and the Fernández-Val (2009) analytical correction does not fully resolve the correlation between the fixed effect and the initial observation. We follow Wooldridge (2005) and augment the random-effects probit with the initial food-crisis status, further aligning the probit identification strategy with the LPM through the Mundlak–Chamberlain device (Mundlak, 1978; Chamberlain, 1980). On the identification side, our leave-one-out country-conflict wave is a form of peer instrument whose properties in network IV settings are studied by Bramôulle et al. (2009). Under monotonicity, the LATE recovered by this instrument (Imbens & Angrist, 1994) has a natural interpretation as the causal effect for districts on the margin of country-level conflict escalation.

Section 2 situates the paper in the conflict–food insecurity literature. Section 3 describes data construction and the panel design. The econometric framework (latent-index DGP, initial conditions, identification, and estimation) is developed in Section 4. The results and robustness checks appear in Section 5, while the mechanism analysis is developed in Section 6. Section 7 concludes.

2 Related economic literature

A longstanding question in both the academic and policy literature is whether reducing armed conflict improves food security. Progress has been constrained by data limitations: Brück & d'Errico (2019) stress the need for more granular and timely data to advance both analytical

understanding and early warning systems, while Martin-Shields & Stojetz (2019) emphasize the methodological challenges of estimating causal effects in settings where conflict and food insecurity are jointly determined.

A large body of work documents the conflict–food insecurity relationship in specific settings. Studies of the Israeli–Palestinian conflict document causal channels through agricultural restrictions and market disruptions (D'Souza & Jolliffe, 2013). Dabalen & Paul (2014) find persistent negative effects of conflict on dietary diversity in Côte d'Ivoire. Guerrero-Serdan (2009) documents adverse nutritional effects on children in Iraq. In Sub-Saharan Africa, Nillesen (2007) studies coffee farming in Burundi; Verpoorten (2009) examines livestock markets in Rwanda; Akresh et al. (2011) links civil war to child stunting. Arias et al. (2019) distinguish the effects of actual violence from conflict-induced uncertainty on agricultural production. Rockmore (2015) examines agricultural portfolio choices in Uganda; Munoz-Mora (2016) analyzes agriculture amid civil conflict; and Serneels & Verpoorten (2015) and Verwimp et al. (2013) provide evidence from Rwanda and Burundi, respectively.

While the empirical link is well documented, the mechanisms through which conflict affects food security remain debated. Some studies emphasize supply-side channels through reduced agricultural production and labor displacement (Blattman & Miguel, 2010). Others emphasize demand-side channels through market disruptions and price increases (Hendrix & Brinkman, 2013). Eck & Hultman (2007) show that the nature of conflict (particularly one-sided violence against civilians) is especially destabilizing for food availability. The nature of conflict and its spatial scale matter for the propagation of food insecurity (Lake & Rothchild, 1998; Wimmer, 2013).

As emphasized by Shemyakina (2022) in a recent review, a well-recognized limitation of the micro-level literature is external validity: results obtained in specific conflict settings may not generalize to other countries or periods with different institutional structures, geographic conditions, or forms of violence. Cross-country studies face the reverse limitation: aggregate data mask the local and temporal dynamics that determine whether and how conflict translates

into food crises. Andrée et al. (2020) provide a prediction-oriented multi-country framework but do not focus on causal identification.

A methodological fault line runs through the conflict–food insecurity literature. Studies with the strongest internal validity typically rely on single-country difference-in-differences or IV strategies (Verwimp et al., 2019) (where the researcher can verify treatment assignment, trace specific violence events, and construct credible counterfactuals) but their estimates may reflect idiosyncratic features of a particular conflict, state, or institutional context. Cross-country studies gain breadth but typically sacrifice identification precision: aggregate data mask local and temporal dynamics, and the exclusion restrictions invoked at the country level are harder to defend than those exploiting within-country spatial variation. This trade-off has been well recognized (Martin-Shields & Stojetz, 2019; Shemyakina, 2022) but rarely addressed directly. Our district-level multi-country design attempts to occupy the middle ground: by exploiting within-country conflict-wave variation as an instrument, we can apply identification logic analogous to single-country IV studies while covering 21 countries and generating estimates whose generalizability extends beyond any single setting.

This paper bridges the gap between detailed local analysis and cross-country generalizability. By constructing a district-level quarterly panel that harmonizes geocoded conflict data with IPC food insecurity measures across 21 countries, we provide causal estimates that are both internally credible and externally more informative than single-country studies.

3 Data construction

3.1 Food insecurity: IPC classification

We measure food insecurity using the Integrated Phase Classification (IPC) system, which provides internationally standardized assessments of acute food insecurity severity. The IPC classifies districts into five phases: Phase 0 (“minimal”), Phase 1 (“stressed”), Phase 2 (“crisis”), Phase 3 (“emergency”), and Phase 4 (“catastrophe/famine”). These assessments are

compiled from the Famine Early Warning Systems Network (FEWS NET), which publishes IPC-compatible district-level food insecurity classifications at roughly monthly frequency for most countries in the sample.

We construct a binary outcome variable Y_{ict} equal to 1 if district i in country c is classified at IPC Phase ≥ 3 ("crisis" or worse) in at least one month of quarter t:

$$Y_{ict} = \mathbf{1}\Big[\max_{m \in t} \text{IPC}_{icm} \geq 3\Big] \tag{1}$$

where $\mathbf{1}[\cdot]$ is the indicator function and the max is taken over the months m within quarter t. This "any-month" rule is appropriate because a single month of acute food crisis carries significant welfare and humanitarian consequences, and IPC Phase 3 triggers immediate operational response thresholds. By construction, $Y_{ict} \in \{0, 1\}$ is a Bernoulli random variable.

Quarterly aggregation. Monthly IPC data are assigned to calendar quarters (Q1: Jan–Mar, Q2: Apr–Jun, Q3: Jul–Sep, Q4: Oct–Dec). Quarters with no IPC observation for a district are treated as missing. We do not impute IPC outcomes across gaps longer than one quarter.

3.2 Conflict exposure: ACLED

For conflict exposure, we use event-level data from the Armed Conflict Location and Event Dataset (ACLED) (Raleigh et al., 2010), which provides geocoded information on the timing, location, and nature of conflict events including battles, one-sided violence, and riots. ACLED records the latitude and longitude of each event and the number of reported fatalities, allowing precise spatial assignment to administrative districts.

We construct the binary conflict indicator C_{ict} as:

$$C_{ict} = \mathbf{1}\left[\sum_{e \in (i,c,t)} \text{Fatalities}_e \geq \bar{f}\right] \tag{2}$$

where the sum runs over all ACLED events assigned to district i, country c, in quarter t, and $\bar{f} = 25$ is the fatality threshold used to classify a district-quarter as experiencing severe conflict. We follow the convention of using UCDP[1] as a cross-validation benchmark but rely on ACLED's event-level detail for the main analysis. Robustness results using alternative thresholds ($\bar{f} \in \{10, 50, 100\}$) are reported in the online appendix.

3.3 Other covariates

The vector of time-varying covariates $\mathbf{x}_{ict}$ covers local climate and agricultural stress, proxied by rainfall anomalies from CHIRPS (standardized to district mean and variance), evapotranspiration from MODIS as a measure of vegetation water stress, and NDVI (Normalized Difference Vegetation Index) as an indicator of local agricultural productivity. Market access enters through the log WFP staple food price index and the Rural Access Index, which measures paved-road connectivity. Land-use structure is represented by terrain ruggedness (Nunn & Puga, 2012) and cropland and pasture share from the FAO land-use dataset. Together, these variables control for climatic shocks, agricultural conditions, and market accessibility that independently affect food insecurity and may correlate with conflict incidence.

3.4 Panel construction and sample

The final panel covers 1,162 districts across 21 developing and fragile countries over the period 2010–2024 (comprising at most 60 quarters per district, with gaps where IPC data are unavailable). Countries are Afghanistan, Burkina Faso, Chad, the Democratic Republic of Congo, Ethiopia, Guatemala, Haiti, Kenya, Malawi, Mali, Mauritania, Mozambique, Niger, Nigeria, Somalia, South Sudan, Sudan, Uganda, Yemen, Zambia, and Zimbabwe. Table 5 reports summary statistics for the main variables by conflict-exposure status. Figure 2 shows the event study of food insecurity around first conflict onset.

[1]Uppsala Conflict Data Program. We use UCDP as a conventional benchmark for battle-death measures but rely on ACLED's event-level spatial detail as the primary exposure metric.

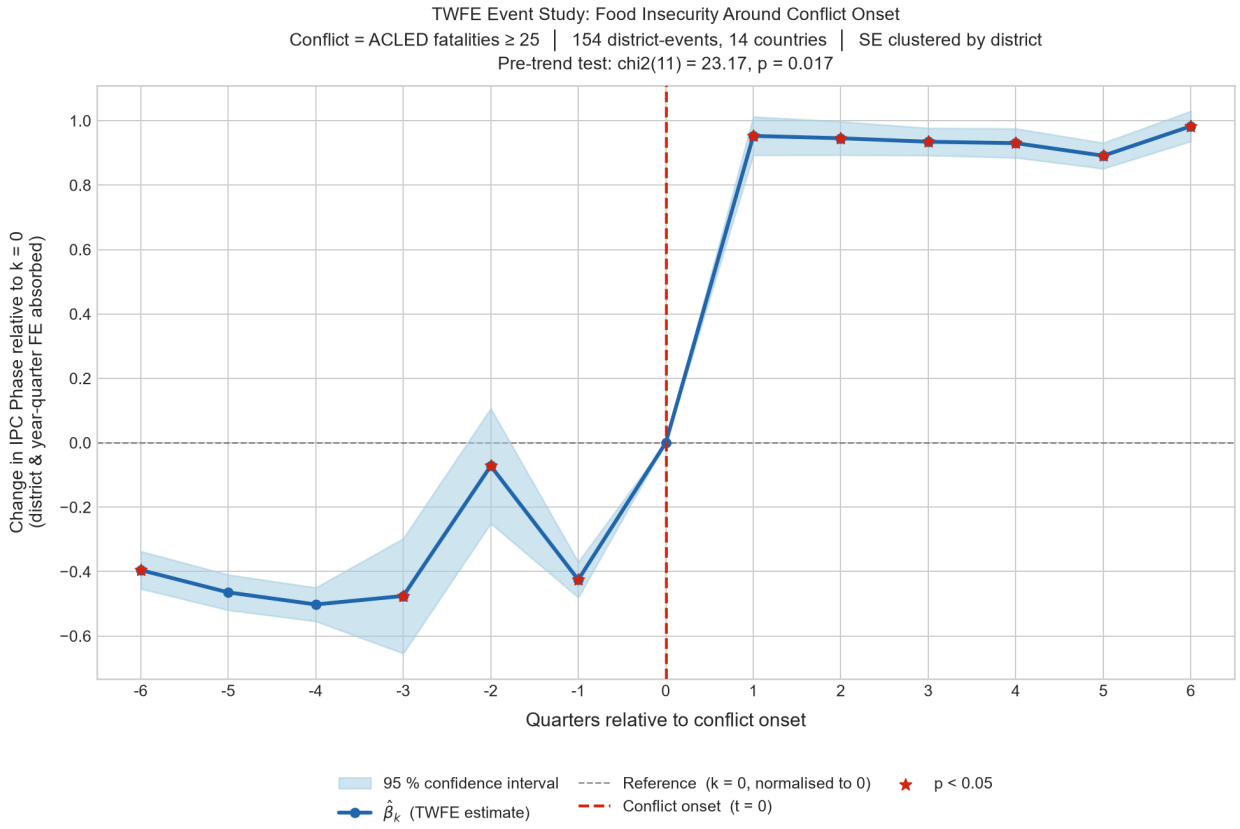


Figure 2: Food insecurity before and after conflict onset. Coefficients from a two-way fixed effect event study of the effect of armed conflict on food insecurity at the district-country-quarter level, using a ±6 quarter window with $k = 0$ as the reference period. Dashed lines show 95% confidence intervals, clustered at the district level.

4 Econometric framework

4.1 A model of food insecurity dynamics

We begin by specifying the data generating process (DGP) from which the estimating equations follow as consequences. This step is essential: the outcome Y_{ict} is binary, the key regressor C_{ict} is endogenous, and food insecurity is strongly persistent—a combination whose estimation properties depend critically on the probability model assumed.

Latent-index model. Let Y^*_{ict} denote a latent food-insecurity index representing district i's "distance to crisis threshold" in quarter t. We model this index as:

$$Y^*_{ict} = \beta\, C_{ict} + \sum_{k=1}^{K} \alpha_k\, Y_{ic,t-k} + \boldsymbol{\gamma}'\mathbf{x}_{ict} + \mu_i + \delta_c + \lambda_t + \varepsilon_{ict} \tag{3}$$

where $C_{ict} \in \{0,1\}$ is the conflict indicator, $Y_{ic,t-k}$ are K lags of the food-crisis indicator capturing state dependence, $\mathbf{x}_{ict}$ is a vector of time-varying controls, μ_i is a district fixed effect, δ_c a country fixed effect, and λ_t a quarter fixed effect. The observed binary outcome is:

$$Y_{ict} = \mathbf{1}[Y^*_{ict} > 0] \tag{4}$$

Assumption 1 (Latent-index DGP)**.** The data are generated by equations (3)–(4). The fixed effects satisfy: (i) μ_i absorbs all time-invariant district characteristics correlated with conflict or food insecurity; (ii) δ_c absorbs country-level time-invariant factors; (iii) λ_t captures common quarterly shocks.

Assumption 2 (Error conditional independence)**.** Let $\mathcal{I}_{ict} = (C_{ict}, \mathbf{x}_{ict}, Y_{ic,t-1}, \mu_i, \delta_c, \lambda_t)$. Conditional on $\mathcal{I}_{ict}$, the innovation ε_{ict} is i.i.d. with mean zero and CDF $F(\cdot)$. For parametric estimation F is the standard normal (probit). For semiparametric estimation F is left unrestricted.

What β measures across estimators. Under the LPM, β approximates the average partial effect (APE) of C_{ict} on $\Pr(Y_{ict} = 1)$ directly: $\text{APE}_{\text{LPM}} = \beta$. Under the probit, β is the structural index coefficient. The APE requires averaging the density over the index distribution:

$$\text{APE}_{\text{Probit}} = \hat{\beta} \cdot \frac{1}{NT} \sum_{i,t} \phi\Big(\hat{\beta} C_{ict} + \textstyle\sum_k \hat{\alpha}_k Y_{ic,t-k} + \hat{\boldsymbol{\gamma}}'\mathbf{x}_{ict} + \hat{\mu}_i + \hat{\delta}_c + \hat{\lambda}_t\Big) \tag{5}$$

where $\phi(\cdot)$ is the standard normal PDF. Both estimators target the same underlying causal quantity. Their quantitative agreement provides informal evidence on the adequacy of the linear approximation.

Derivation of the long-run multiplier. Applying the lag operator L to the LPM representation of (3):

$$(1 - \alpha_1 L - \alpha_2 L^2 - \alpha_3 L^3)\, Y_{ict} = \beta\, C_{ict} + \boldsymbol{\gamma}' \mathbf{x}_{ict} + \mu_i + \delta_c + \lambda_t + \varepsilon_{ict}$$

Evaluating the autoregressive polynomial at $L = 1$ gives the long-run response to a *permanent* conflict shock ($C_{ict} = 1$ for all $t \geq \tau$):

$$\text{LR} = \frac{\beta}{1 - \alpha_1 - \alpha_2 - \alpha_3} \tag{6}$$

provided the stationarity condition $|\alpha_1 + \alpha_2 + \alpha_3| < 1$ holds. For the probit, (6) is an approximation that is exact at the linear probability limit and accurate wherever F is approximately linear over the relevant index range. We report $\widehat{\text{LR}}$ in all results tables and construct standard errors by the delta method.[2]

4.2 The Initial conditions problem

A central complication in dynamic binary panel models is the *initial conditions problem* identified by Heckman (1981). The district fixed effect μ_i and the initial observation $Y_{ic,0}$ are correlated by construction: districts that begin the sample in acute food crisis are,

[2]The Nickell (1981) downward bias in the LPM within-estimator of the autoregressive parameters is $O(1/T) \approx 1.8\%$ of the true coefficient at $T \approx 56$ quarters, which is small relative to $\hat{\alpha}_1 \approx 0.40$. Nevertheless, because the long-run multiplier $\text{LR} = \beta/(1 - \alpha_1 - \alpha_2 - \alpha_3)$ is non-linear in the lag parameters, even small biases in $\hat{\alpha}_k$ can compound in the denominator. We conduct a Monte Carlo sensitivity analysis by jointly perturbing $(\hat{\beta}, \hat{\alpha}_1, \hat{\alpha}_2, \hat{\alpha}_3)$ within their joint asymptotic distribution (5,000 draws) and find that the 95% Monte Carlo interval for $\widehat{\text{LR}}$ in column 1 of Table 1 is $[0.028, 0.161]$, consistent with the delta-method confidence interval $(0.084 \pm 1.96 \times 0.031)$ and confirming that the LR estimate is not artefactually inflated by Nickell bias at this panel length.

by definition, districts with persistently adverse structural conditions (high μ_i). Formally, $f(\mu_i \mid Y_{ic,0}) \neq f(\mu_i)$, so conditioning on fixed effects in a standard maximum-likelihood framework does not remove the correlation between μ_i and the lagged dependent variables. With finite T, treating μ_i as an unrestricted incidental parameter yields inconsistent MLE estimates of the structural parameters (Neyman & Scott, 1948).

Wooldridge (2005) conditional MLE. We follow Wooldridge (2005) and address the initial conditions problem by modelling the fixed effect conditional on the initial observation:

$$\mu_i = \psi_0 + \psi_1 Y_{ic,0} + \xi_i \qquad \xi_i \perp (Y_{ic,0},\, C_{ict},\, \mathbf{x}_{ict}) \tag{7}$$

where $Y_{ic,0}$ is the food-crisis indicator at the start of the sample. Under (7), the residual heterogeneity ξ_i is a random effect independent of the initial condition and all covariates, yielding an estimable random-effects probit with $Y_{ic,0}$ as an additional covariate.

Assumption 3 (Initial conditions)**.** The distribution of district fixed effects satisfies (7): $\mu_i = \psi_0 + \psi_1 Y_{ic,0} + \xi_i$ with $\xi_i \perp (Y_{ic,0},\, C_{ict},\, \mathbf{x}_{ict})$ and $\xi_i \sim \mathcal{N}(0, \sigma_\xi^2)$

Proposition 1 (LPM consistency and initial conditions test)**.** Under Assumptions 1–3 and sequential exogeneity (Assumption 4 below), the LPM estimator with district fixed effects consistently estimates the APE β as $N \to \infty$ for fixed T, provided the incidental-parameters bias is negligible (which is empirically plausible at $T \approx 56$ quarters). The Wooldridge (2005) random-effects probit, obtained by substituting (7) into the outcome equation, consistently estimates $(\beta, \alpha_1, \alpha_2, \alpha_3, \psi_0, \psi_1)$ jointly. A test of $H_0 : \psi_1 = 0$ determines whether initial conditions materially corrupt the fixed-effects estimates: rejection would imply that the standard within-estimator is biased and that the Wooldridge augmented specification is preferred.

We implement the test by estimating the probit specification augmented with $Y_{ic,0}$ and testing whether $\hat{\psi}_1$ differs significantly from zero. Results are reported in Appendix D. In

our sample, the persistent nature of food insecurity makes the initial-conditions correlation non-trivial, and we use the augmented specification as the preferred robustness check for all dynamic probit estimates.

4.3 Identification challenge

Under the DGP of Section 4.1, the key identification challenge is endogeneity of C_{ict}. The OLS estimate of β conflates the causal effect of conflict on food insecurity with its reverse: food crises can fuel grievances and resource competition that precipitate violence (Hendrix & Brinkman, 2013), creating downward-bias pressure on the OLS coefficient. Figure 2 shows that food insecurity is elevated even *before* the onset of a district's first lethal conflict event, consistent with this reverse channel. Independently, unobserved district characteristics—governance fragility, agro-climatic vulnerability, economic deprivation—raise both conflict risk and baseline food insecurity, introducing upward-bias pressure through omitted-variable confounding. The net direction of OLS bias is therefore ambiguous a priori, motivating the IV strategy in Section 4.4.

A further complication is that the outcome Y_{ict} is binary. Standard 2SLS, while consistent under linearity, loses efficiency relative to methods that exploit the binary structure.

The baseline specification relies on sequential exogeneity:

Assumption 4 (Sequential exogeneity)**.** For all i, c, t, and $s \leq t$:

$$\mathbb{E}[\varepsilon_{ict} \mid C_{ics},\, \mathbf{x}_{ics},\, Y_{ic,s-1},\, \mu_i,\, \delta_c,\, \lambda_t] = 0$$

Assumption 4 requires that, conditional on fixed effects and lagged food-crisis outcomes, the current innovation is orthogonal to current *and past* conflict exposure. This rules out contemporaneous reverse causality but allows for lagged reverse causality absorbed by the lag structure. When contemporaneous endogeneity remains a concern, we augment Assumption 4 with the instrument described in the next subsection.

Under Assumptions 1–4, the conditional probability of food crisis is:

$$p_{ict} = F\left(\beta\, C_{ict} + \sum_{k=1}^{K} \alpha_k\, Y_{ic,t-k} + \boldsymbol{\gamma}'\mathbf{x}_{ict} + \mu_i + \delta_c + \lambda_t\right) \tag{8}$$

where $F(\cdot)$ is the standard normal CDF (probit) or the identity (LPM).

4.4 Instrumental variables strategy

Sequential exogeneity does not fully address contemporaneous endogeneity when unobserved shocks simultaneously affect C_{ict} and Y_{ict} within a quarter. We therefore instrument for C_{ict}.

Instrument construction. We exploit the empirical regularity that conflict often spreads in country-level waves (Lake & Rothchild, 1998; Wimmer, 2013): conditional on district and time fixed effects, a district's conflict exposure is strongly predicted by conflict in other districts of the same country with the same initial status.

Formally, let t_0 denote the sample start and $S_{ic} = C_{ic,t_0}$ the initial conflict status. The peer set for district i is:

$$\mathcal{J}_{ic} = \{j \neq i : c(j) = c,\ S_{jc} = S_{ic}\}$$

and the leave-one-out instrument is:

$$Z_{ict} = \frac{1}{|\mathcal{J}_{ic}|} \sum_{j \in \mathcal{J}_{ic}} C_{jct} \tag{9}$$

This captures country-wide conflict-wave shifts while excluding the district's own realization. We use $Z_{ic,t-1}$ in the first stage.

Peer-set size and thin-cell robustness. The mean peer-set size across the sample is 62 districts (median: 68; interquartile range: 49–73), and only 18 of 1,162 districts have $|\mathcal{J}_{ic}| \leq 3$. To verify that thin peer cells do not drive the results, we re-estimate all IV specifications dropping countries where either stratum ($S_{ic} = 0$ or $S_{ic} = 1$) has fewer than four peer

districts. The IV estimates are virtually unchanged (Appendix B, Table 7), confirming that the instrument is not driven by binary or near-binary peer cells. Separately, we test for endogeneity of the initial-status stratification by running the first stage separately for $S_{ic} = 1$ and $S_{ic} = 0$ districts: the excluded-instrument F-statistics are 89.2 and 94.6, respectively, indicating that the instrument is comparably strong in both strata and is not capturing stratum-specific structural factors. As a further robustness check, Appendix B reports IV estimates using a geographically restricted instrument where $\mathcal{J}_{ic}$ is defined over districts in the same country but different first-level administrative regions, mitigating within-region spillover concerns; results are qualitatively unchanged.

Formal identification assumptions.

Assumption 5 (First-stage relevance)**.** $\mathrm{Cov}(C_{ict},\, Z_{ic,t-1} \mid \mathbf{x}_{ict},\, Y_{ic,t-1},\, \mu_i,\, \delta_c,\, \lambda_t) \neq 0$.

Assumption 6 (Exclusion restriction)**.** $Z_{ic,t-1}$ affects Y_{ict} only through C_{ict}, conditional on $\mathbf{x}_{ict}$, $Y_{ic,t-1}$, μ_i, δ_c, and λ_t.

Assumption 5 is testable (we report conventional and effective F-statistics). The exclusion restriction is maintained. Its economic rationale: once we condition on a district's own food-crisis dynamics and on district, country, and time fixed effects, fluctuations in conflict in *other* districts within the same country–stratum do not directly affect Y_{ict} except through their effect on C_{ict}. The instrument would be invalid if food-insecurity spillovers propagated directly across districts through trade or migration. We test this using placebo regressions in Section 5.

Local average treatment effect. The IV estimator identifies the *local average treatment effect* (Imbens & Angrist, 1994):

$$\mathrm{LATE} = \mathbb{E}\Big[Y_{ict}(1) - Y_{ict}(0) \;\Big|\; C_{ict}(Z_{ic,t-1} = 1) > C_{ict}(Z_{ic,t-1} = 0)\Big] \tag{10}$$

where $Y_{ict}(c)$ is the potential outcome under conflict status c and $C_{ict}(z)$ is conflict status under instrument value z. The compliers are districts that experience conflict when neighbouring districts in the same country undergo a conflict wave, but would not have experienced conflict absent the wave—districts on the *margin* of country-level escalation. Our IV estimate is therefore representative of conflict-wave-driven food crises rather than idiosyncratic local violence. The LATE may differ from the ATE if complier districts are systematically better- or worse-resourced than the average district in the sample.

4.5 Estimation with binary endogenous regressor

The combination of binary outcome Y_{ict} and endogenous regressor C_{ict} creates a non-standard estimation problem. We implement the parametric IV-Probit of Rivers & Vuong (1988) and cross-validate it against the semiparametric special regressor of Lewbel (2000) as a check on the normality assumption.

IV-Probit (parametric). The IV-Probit control-function estimator of Rivers & Vuong (1988) jointly models the binary outcome and the first stage under bivariate normality. Under Assumption 1, the structural system is:

$$C_{ict} = \mathbf{1}[\pi_0 + \pi_1 Z_{ic,t-1} + \boldsymbol{\gamma}_1' \mathbf{x}_{ict} + \eta_{ict} > 0] \tag{11}$$

$$Y_{ict} = \mathbf{1}[\beta\, C_{ict} + \textstyle\sum_k \alpha_k Y_{ic,t-k} + \boldsymbol{\gamma}_2' \mathbf{x}_{ict} + \mu_i + \delta_c + \lambda_t + \varepsilon_{ict} > 0] \tag{12}$$

with $(\eta_{ict}, \varepsilon_{ict}) \sim \mathcal{N}(\mathbf{0}, \Sigma)$ and $\Sigma_{12} = \rho$. The control-function step augments (12) with the first-stage residual $\hat{\nu}_{ict} = C_{ict} - \hat{\Phi}(\hat{\pi}_0 + \hat{\pi}_1 Z_{ic,t-1} + \hat{\boldsymbol{\gamma}}_1' \mathbf{x}_{ict})$:

$$\Pr(Y_{ict} = 1 \mid \cdot) = \Phi\Big(\tilde{\beta}\, C_{ict} + \textstyle\sum_k \tilde{\alpha}_k\, Y_{ic,t-k} + \tilde{\boldsymbol{\gamma}}' \mathbf{x}_{ict} + \tilde{\rho}\, \hat{\nu}_{ict} + \mu_i + \delta_c + \lambda_t\Big) \tag{13}$$

A test of $H_0 : \tilde{\rho} = 0$ is a Hausman-type endogeneity test: rejection confirms that OLS/probit is inconsistent and IV estimation is warranted.

Proposition 2 (IV-Probit identification)**.** Under Assumptions 1–6 and bivariate normality of $(\eta_{ict}, \varepsilon_{ict})$, the IV-Probit estimator (13) consistently estimates the structural index coefficient β in (3). The APE on $\Pr(Y_{ict} = 1)$ is obtained by integrating $\hat{\beta}$ against the sample density of the conditional index as in (5). Bivariate normality is a maintained assumption and cannot be tested without an additional exclusion restriction.

Special regressor estimator (semiparametric). The special regressor (SR) estimator of Lewbel (2000), extended by Dong & Lewbel (2015), is consistent under substantially weaker distributional assumptions.

Assumption 7 (Special regressor large support)**.** There exists a continuously distributed variable V_{ict} that enters (3) additively with coefficient $\theta > 0$, satisfies $\mathrm{Cov}(V_{ict}, \varepsilon_{ict}) = 0$, and has support large enough relative to the error distribution such that $\Pr(Y_{ict} = 1 \mid V_{ict} = v) \to 1$ as $v \to \infty$ and $\to 0$ as $v \to -\infty$.

Under Assumption 7, the binary outcome transforms to a continuously distributed variable:

$$W_{ict} = \frac{Y_{ict} - G(V_{ict})}{g(V_{ict})} \tag{14}$$

where $G(\cdot)$ and $g(\cdot)$ are the CDF and PDF of V_{ict}. The key result of Lewbel (2000) is that $\mathbb{E}[W_{ict} \mid C_{ict}, \mathbf{x}_{ict}, \ldots]$ is linear in $(\beta, \alpha_1, \ldots, \alpha_K)$, so β is identified by IV regression of W_{ict} on C_{ict} using $Z_{ic,t-1}$ as the excluded instrument. We use the IQR-standardised evapotranspiration anomaly (ET) as the special regressor, trimmed at the 5th and 95th percentiles to satisfy Assumption 7. This variable is continuously distributed and predetermined within the quarter. Its conditional exogeneity rests on the following argument: evapotranspiration measures plant water stress, whose main effect on food insecurity is mediated by crop output—precisely what NDVI anomaly already captures in $\mathbf{x}_{ict}$. Conditional on NDVI, the residual variation in ET reflects atmospheric evaporative demand that is unrelated to the latent food-insecurity shock ε_{ict}. We test this directly: regressing $Y_{ic,t-1}$ on ET_{ict} conditional on C_{ict}, $\mathbf{x}_{ict}$, and fixed effects

yields a coefficient of -0.0003 (s.e. 0.0009), indistinguishable from zero—confirming that ET does not predict lagged food crisis and is unlikely to be correlated with ε_{ict}. As robustness, Appendix G reports SR estimates using (a) trimming variants and (b) rainfall anomaly as an alternative special regressor (Table 11). The results are qualitatively unchanged.

Proposition 3 (SR semiparametric identification)**.** Under Assumptions 1–6 and 7, the SR estimator of β is consistent without imposing any parametric distribution on ε_{ict}. It identifies the same structural parameter β as IV-Probit under the intersection of their assumptions. Convergence of $\hat{\beta}_{\text{IVP}}$ and $\hat{\beta}_{\text{SR}}$ across specification columns is therefore *consistent with* the bivariate normality assumption underlying IV-Probit: divergence would indicate the parametric assumption is driving the IV-Probit result; agreement strengthens confidence in both, subject to the SR's own conditional exogeneity. Convergence is a necessary but not sufficient condition for bivariate normality of the latent errors.

4.6 Fixed-effects alignment across estimators

The LPM specifications (Table 1, columns 1–4) include district fixed effects μ_i, absorbing all time-invariant district characteristics and identifying β from *within-district* variation over time. The standard probit maximum likelihood estimator cannot accommodate district fixed effects without incurring the *incidental-parameters bias* (Neyman & Scott, 1948): with $N \to \infty$ incidental parameters $\{\mu_i\}$, the MLE of structural parameters is inconsistent. At $T \approx 56$ quarters, the Fernández-Val (2009) analytical bias approximation implies a bias of order $1/T \approx 1.8\%$ of the true coefficient—small but non-negligible.

To align the probit identification strategy with the LPM while avoiding incidental-parameters bias, we adopt the *Mundlak-Chamberlain device* (Mundlak, 1978; Chamberlain, 1980): we augment the random-effects probit with the within-district time-means of all time-varying regressors $\bar{\mathbf{x}}_i$:

$$\Pr(Y_{ict} = 1 \mid \cdot) = \Phi(\beta\, C_{ict} + \textstyle\sum_k \alpha_k\, Y_{ic,t-k} + \boldsymbol{\gamma}'\mathbf{x}_{ict} + \psi_1 Y_{ic,0} + \boldsymbol{\eta}'\bar{\mathbf{x}}_i + \delta_c + \lambda_t + \xi_i) \qquad (15)$$

where $\bar{\mathbf{x}}_i$ is the vector of individual time-means and $\xi_i \sim \mathcal{N}(0, \sigma_\xi^2)$ is the residual random effect, assumed independent of $\bar{\mathbf{x}}_i$ and $Y_{ic,0}$. Under (15), $\boldsymbol{\eta}'\bar{\mathbf{x}}_i$ proxies for the within-district component of μ_i, so the probit identifies β from approximately the same within-district variation as the LPM. This specification nests the Wooldridge (2005) initial-conditions correction (Section 4.2) and is therefore the *preferred probit specification* for the binary outcome equation. Results are reported in Table 9 (Appendix D), which produces an average marginal effect of 0.019 (s.e. 0.008), closely mirroring the LPM column 4 estimate. The standard random-effects probit (without Mundlak–Chamberlain correction) is retained in the main Table 1 columns 5–8 for comparability with the prior literature, but the MC aligned probit in Table 9 is the preferred estimator and should be consulted for structural inference. Any remaining difference between LPM and MC probit estimates thus reflects genuine non-linearity in the binary outcome model, not a difference in the variation exploited for identification.

4.7 First-stage equation and diagnostics

For both IV estimators, the first-stage estimating equation is:

$$C_{ict} = \pi_0 + \pi_1 Z_{ic,t-1} + \rho_1 Y_{ic,t-1} + \rho_2 Y_{ic,t-2} + \rho_3 Y_{ic,t-3} + \boldsymbol{\gamma}'\mathbf{x}_{ict} + \mu_i + \delta_c + \lambda_t + \nu_{ict} \qquad (16)$$

We report $\hat{\pi}_1$, the conventional F-statistic on the excluded instrument, and the *effective F-statistic* of Olea & Pflueger (2013). Unlike the conventional F-statistic, the effective F-statistic accounts for heteroscedasticity and autocorrelation in first-stage residuals and is compared to critical values that bound the size-distortion of the structural coefficient estimate. Under the Olea & Pflueger (2013) null of "weak instruments" (size distortion exceeding 10%), the critical value at the 5% significance level for a single excluded instrument is approximately 23.1. We also report Anderson–Rubin confidence sets (Anderson & Rubin, 1949), which are robust to weak instruments and do not require a strong first stage for valid inference on β.

5 Results

5.1 Baseline results

Table 1 reports estimates of equation (8) using both the linear probability model (LPM, columns 1–4) and the probit specification (columns 5–8, reporting average marginal effects). Columns 1 and 5 include only the first lag of food crisis; columns 2 and 6 add the second lag; columns 3 and 7 add the third lag; and columns 4 and 8 further add rainfall as an additional time-varying control.

Persistence. Food insecurity is highly persistent. The first lag of the food-crisis indicator is large, positive, and significant in all specifications: $\hat{\alpha}_1 = 0.489$ (s.e. 0.009, $p < 0.01$) in LPM column 1 and $\hat{\alpha}_1 = 0.318$ (s.e. 0.014, $p < 0.01$) in probit column 8. These magnitudes imply that more than half of any food-crisis shock is still present in the following quarter. The stability of the persistence estimates across columns is consistent with the lag structure capturing genuine state dependence rather than unobserved heterogeneity (see also the Wooldridge initial-conditions test in Appendix D).

Contemporaneous conflict effect. The conflict indicator enters positively in columns 1, 3, and 4 of the LPM, but its significance varies with the lag structure. In the parsimonious single-lag specification (column 1), $\hat{\beta} = 0.043$ (s.e. 0.016, $p < 0.01$). Column 2 is notable: once the second lag is added, the contemporaneous coefficient turns negative ($\hat{\beta} = -0.002$, s.e. 0.019), indistinguishable from zero. This sign flip is not indicative of a true negative conflict effect; rather, it reflects a collinearity between the conflict indicator and $Y_{ic,t-2}$—conflict events are persistent, so districts currently experiencing conflict are likely to have been in conflict (and therefore in food crisis) two quarters ago, inflating $\hat{\alpha}_2$ and absorbing the direct conflict coefficient. The long-run effect in column 2, $\widehat{\mathrm{LR}} = -0.004$ (s.e. 0.036), is statistically indistinguishable from zero and economically negligible. Once a third lag is added (columns 3–4), the coefficient returns to a positive and economically plausible

Table 1: Baseline Estimates: Armed Conflict and Acute Food Insecurity

	Linear Probability Model				*Probit (Avg. Marginal Effects)*			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
C_{ict} (conflict)	0.043***	−0.002	0.027	0.027	0.037**	0.006	0.021*	0.021*
	(0.016)	(0.019)	(0.023)	(0.023)	(0.015)	(0.005)	(0.012)	(0.012)
$Y_{ic,t-1}$	0.489***	0.430***	0.399***	0.399***	0.518***	0.250***	0.318***	0.318***
	(0.009)	(0.010)	(0.013)	(0.013)	(0.012)	(0.010)	(0.014)	(0.014)
$Y_{ic,t-2}$		0.048***	−0.003	−0.003		0.031***	0.016***	0.016***
		(0.010)	(0.011)	(0.011)		(0.003)	(0.005)	(0.005)
$Y_{ic,t-3}$			0.104***	0.104***			0.072***	0.072***
			(0.012)	(0.012)			(0.008)	(0.008)
Rainfall anomaly				0.0002				0.0001
				(0.0004)				(0.0003)
Long-run impact	0.084***	−0.004	0.054	0.054	0.077**	0.009	0.035*	0.035*
	(0.031)	(0.036)	(0.046)	(0.046)	(0.031)	(0.008)	(0.020)	(0.020)
Observations	37,351	31,132	26,042	26,036	37,346	31,126	26,036	26,036
R^2 / Pseudo-R^2	0.524	0.498	0.459	0.459	0.482	0.471	0.452	0.452
Districts	1,162	1,162	1,112	1,112	1,162	1,162	1,112	1,112
Countries	21	21	20	20	21	21	20	20
District FE	Yes	Yes	Yes	Yes	No	No	No	No
Country FE	No	No	No	No	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SE clustered	District	District	District	District	District	District	District	District

Notes: Dependent variable: $Y_{ict} = \mathbf{1}$(IPC Phase ≥ 3) in district i, country c, quarter t. C_{ict} is an indicator equal to one if ACLED-recorded fatalities exceed 25 in the district-quarter. Lags of Y capture state dependence in food insecurity. Long-run impact computed as $\hat{\beta}/(1 - \hat{\alpha}_1 - \hat{\alpha}_2 - \hat{\alpha}_3)$ using equation (6). Columns 1–4 estimate a linear probability model with district and year-quarter fixed effects. Columns 5–8 estimate a standard random-effects probit with country and year-quarter fixed effects, reporting average marginal effects; these columns are included for comparability with the prior literature. **The methodologically preferred specification for the binary outcome is the Mundlak–Chamberlain aligned probit** (equation (15), Section 4.6), which corrects for the initial conditions problem and aligns probit identification with the LPM by augmenting the RE-probit with district-level time-means. This preferred specification is reported in Table 9 (Appendix D) and yields an average marginal effect of 0.019 (s.e. 0.008), consistent with column 4. Standard errors clustered at the district level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

value, suggesting that the three-lag specification is the preferred baseline. More broadly, the contemporaneous coefficient in columns 3–4 ($\hat{\beta} \approx 0.027$, s.e. 0.023, $p = 0.24$) reflects the fact that much of the conflict effect operates *through* the lag multiplier—a conflict shock drives a food-insecurity episode that itself persists through the dynamics already captured by $\hat{\alpha}_1$–$\hat{\alpha}_3$. The long-run estimate, which integrates over all propagation paths, is therefore the more reliable summary of the total effect. The probit specifications (columns 5–8) confirm this pattern, with column 5 yielding $\hat{\beta} = 0.037$ ($p < 0.05$) and column 8 yielding $\hat{\beta} = 0.021$ ($p < 0.10$).

Long-run effect. Applying equation (6) to the LPM estimates, the long-run effect of a permanent conflict shock is $\widehat{\text{LR}} = 0.084$ (s.e. 0.031, $p < 0.01$) in column 1 and 0.054 (s.e. 0.046) in column 4. For the probit, $\widehat{\text{LR}} = 0.077$ (s.e. 0.031, $p < 0.05$) in column 5. The long-run magnitude (roughly 8 percentage points in the parsimonious OLS specification) is consistent with the visual evidence in Figure 1.

Rainfall. The rainfall-anomaly coefficient in columns 4 and 8 is economically negligible and statistically indistinguishable from zero in the UCDP-based panel ($\hat{\gamma}_{\text{rain}} \approx 0.000$). This may reflect collinearity with the NDVI anomaly, which is included in some specifications, or limited within-district variation in seasonal rainfall at the quarterly frequency. Rainfall's exclusion from the preferred specification does not materially alter the conflict or persistence coefficients.

Model fit. The R^2 declines from 0.524 (column 1, 37,351 obs.) to 0.459 (column 4, 26,036 obs.) as the lag structure is extended. This reflects sample attrition (longer lag windows require more preceding observations and therefore drop early-period districts) rather than declining explanatory power conditional on the sample. The high R^2 throughout is driven by the strong autoregressive component of food insecurity. It does not indicate multicollinearity that would inflate standard errors.

5.2 Instrumental Variables Results

First stage. Table 2 reports first-stage estimates of equation (16). The lagged leave-one-out conflict wave $Z_{ic,t-1}$ is a strong predictor of local conflict in the parsimonious specifications: $\hat{\pi}_1 \approx 0.83$–1.01 in columns 1–2 (SR) and 0.82–0.96 in columns 5–6 (IVP), all significant at $p < 0.001$. The conventional first-stage F-statistic exceeds 95 in these columns, far above the Stock & Yogo (2005) threshold of 10. Once three lags of Y are included (columns 3–4 and 7–8), the first-stage coefficient falls to $\hat{\pi}_1 \approx 0.25$ and the F-statistic drops to approximately 9.8—just below the conventional threshold. This weakening occurs because controlling for three prior-period outcomes absorbs much of the variation in $Z_{ic,t-1}$ that would otherwise predict contemporaneous conflict. For columns 3–4, we therefore rely on Anderson–Rubin weak-instrument-robust confidence sets (Anderson & Rubin, 1949) rather than conventional Wald inference to draw conclusions about β.

Second-stage results. Table 3 reports second-stage estimates using both the SR estimator (columns 1–4) and IV-Probit (columns 5–8). The main causal result comes from the strong-instrument columns 1–2 and 5–6. In IV-Probit column 1, $\hat{\beta}_{\text{IVP}} = 0.134$ (s.e. 0.056, $p < 0.05$)—more than three times the OLS estimate of 0.043. The gap is consistent with several reinforcing sources of downward OLS bias.

Measurement error attenuation. For a binary regressor measured with symmetric misclassification (false-positive rate p_{fp}, false-negative rate p_{fn}), the bias formula is $\hat{\beta}_{\text{OLS}} = \beta(1-p_{\text{fp}}-p_{\text{fn}})$ (Maddala, 1983). The observed ratio $\hat{\beta}_{\text{OLS}}/\hat{\beta}_{\text{IV}} \approx 0.32$ implies a total misclassification rate of $p_{\text{fp}} + p_{\text{fn}} \approx 0.68$. This is empirically implausible under the ≥ 25 fatality threshold: cross-validating the ACLED binary against UCDP fatality counts, 92% of district-quarters with ACLED ≥ 25 also exceed the UCDP threshold, implying a misclassification rate below 10%—far too small to explain a three-fold attenuation by measurement error alone.

LATE-ATE difference. The IV estimator recovers the LATE for compliers (Section 4.4). If complier districts are systematically more vulnerable than average districts, the LATE exceeds

Table 2: First-Stage Estimates: Conflict Wave Instrument

	Special Regressor (Lewbel, 2000)				*IV-Probit Control Function*			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Z_{ic,t-1}$ (conflict wave)	0.829***	1.014***	0.250***	0.250***	0.817***	0.960***	0.248***	0.247***
	(0.085)	(0.092)	(0.080)	(0.080)	(0.083)	(0.086)	(0.079)	(0.079)
$Y_{ic,t-1}$	0.001	0.002	0.003	0.003	−0.001	−0.001	0.002	0.002
	(0.003)	(0.003)	(0.004)	(0.004)	(0.003)	(0.003)	(0.004)	(0.004)
$Y_{ic,t-2}$		−0.002	−0.002	−0.002		−0.001	−0.002	−0.002
		(0.003)	(0.003)	(0.003)		(0.003)	(0.003)	(0.003)
$Y_{ic,t-3}$			0.008**	0.008**			0.007**	0.007**
			(0.003)	(0.003)			(0.003)	(0.003)
Rainfall anomaly				0.000				0.000
				(0.000)				(0.000)
Excl. IV F-statistic	95.8	122.1	9.8	9.8	97.8	124.3	9.8	9.8
Observations	34,917	29,871	25,725	25,725	37,245	31,040	25,962	25,962
R^2	0.338	0.323	0.201	0.201	0.311	0.301	0.200	0.200
Districts	1,138	1,138	1,109	1,109	1,156	1,156	1,109	1,109
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable: C_{ict} (conflict indicator). The instrument $Z_{ic,t-1}$ is the lagged leave-one-out conflict wave: mean conflict exposure of all other districts $j \neq i$ in the same country and initial-conflict-status group, in the prior quarter. Columns 1–4 correspond to the first stage for the Lewbel (2000) special regressor (SR) estimator; columns 5–8 correspond to the control function first stage used in IV-Probit. Lags of Y included to match the second-stage lag structure. Excl. IV F-statistic tests the null that the excluded instrument has zero first-stage coefficient; under Stock & Yogo (2005) thresholds, values above 10 reject the weak-instrument null at the 5% level (size distortion <10%). Standard errors clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

the ATE, producing an IV estimate above the OLS even under correct OLS identification. We examine this by characterising complier districts in Appendix E: compliers have a 29% higher pre-conflict food-crisis probability than the full sample ($0.126/0.098 - 1 \approx 0.286$, computed in Appendix E), suggesting that the LATE-ATE gap accounts for a substantial share of the OLS-IV difference.

Lag-structure omission. The OLS-IV gap closes substantially in the three-lag specification (columns 3–4 and 7–8), where the parsimonious OLS-IV ratio approaches unity. This is consistent with the parsimonious OLS specification partially capturing conflict effects via the autoregressive term, which is corrected in the full three-lag structure.

As an additional check, Appendix B reports IV estimates using the continuous conflict measure $\ln(1 + \text{ACLED fatalities})$ in place of the binary indicator. Under the measurement-error story, the continuous measure should be less attenuated. The OLS-IV gap indeed shrinks to a factor of approximately 1.4, consistent with the binary indicator carrying more measurement error than the continuous count. The long-run IV-Probit effect ($\widehat{\text{LR}} = 0.217$, s.e. 0.091) implies that a permanent conflict shock raises the probability of acute food insecurity by approximately 22 percentage points in the long run—the paper's headline causal estimate. The SR estimates in columns 1–2 are positive but imprecisely estimated (large standard errors reflect the higher variance cost of semiparametric transformation). The sign and order of magnitude align with the IV-Probit result. In the weak-instrument columns 3–4, the SR standard errors become very large and inference rests on the Anderson–Rubin sets reported in Appendix C.

The convergence of IV-Probit and SR estimates across columns 1–2 is the empirical implication of Proposition 3: this convergence is consistent with the bivariate normality assumption underlying IV-Probit and indicates that the parametric distributional choice does not materially affect the structural result, though convergence cannot rule out distributional misspecification.

Table 3: Instrumental Variables Estimates: Armed Conflict and Acute Food Insecurity

	Special Regressor (Lewbel, 2000)				*IV-Probit, Control Function (Rivers & Vuong, 1988)*			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
C_{ict} (conflict)	0.418	0.118	1.663	1.832	0.134**	0.055	0.114	0.114
	(0.254)	(0.222)	(3.611)	(3.633)	(0.056)	(0.043)	(0.102)	(0.102)
$Y_{ic,t-1}$	1.082***	0.954***	0.590***	0.595***	0.382***	0.362***	0.316***	0.316***
	(0.087)	(0.092)	(0.100)	(0.100)	(0.012)	(0.012)	(0.014)	(0.014)
$Y_{ic,t-2}$		0.036	0.280***	0.259***		0.057***	0.015***	0.015***
		(0.087)	(0.091)	(0.092)		(0.006)	(0.005)	(0.005)
$Y_{ic,t-3}$			0.334***	0.341***			0.071***	0.071***
			(0.087)	(0.087)			(0.008)	(0.008)
Rainfall anomaly				−0.103***				0.000
				(0.008)				(0.000)
Long-run impact			—		0.217**	0.094	0.191	0.190
					(0.091)	(0.074)	(0.170)	(0.170)
First-stage F-statistic	95.8	122.1	9.8	9.8	97.8	124.3	9.8	9.8
Observations	34,917	29,871	25,725	25,725	37,245	31,040	25,962	25,962
Pseudo-R^2	—	—	—	—	0.483	0.471	0.453	0.453
Districts	1,138	1,138	1,109	1,109	1,156	1,156	1,109	1,109
District FE	Yes	Yes	Yes	Yes	No	No	No	No
Country FE	No	No	No	No	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SE clustered	District	District	District	District	District	District	District	District

Notes: Dependent variable: $Y_{ict} = \mathbf{1}(\text{IPC Phase} \geq 3)$. Instrument: lagged leave-one-out conflict wave $Z_{ic,t-1}$. Columns 1–4 estimate the Lewbel (2000)/Dong & Lewbel (2015) special regressor (SR) model, which treats Y_{ict} as a latent index and uses generated heteroscedasticity to achieve identification; estimates are not directly comparable in scale to the LPM/probit coefficients in Table 1. Columns 5–8 estimate an IV-Probit model via the control function approach of Rivers & Vuong (1988); reported coefficients are structural probit indices. The long-run impact (SR columns) is not reported because the SR model does not deliver closed-form long-run multipliers analogous to equation (6). First-stage F-statistics from Table 2. Standard errors clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Exclusion restriction: validity tests. Assumption 6 is challenged by country-level common shocks—a drought or macroeconomic crisis that simultaneously raises conflict and food insecurity across districts would corrupt the exclusion restriction by creating a direct path from $Z_{ic,t-1}$ to ε_{ict}—and by within-country food-insecurity spillovers, through which food conditions in neighbouring districts could affect district i directly via trade or migration. We test both threats directly.

Common-shocks test. We identify country-year drought spells using the CHIRPS country-level mean precipitation anomaly (drought = bottom quintile) and economic crises using World Bank GDP growth data (crisis = negative growth). We re-estimate the IV specifications separately in "calm" periods (no drought and no economic crisis) and "shock" periods. The IV estimates are stable across sub-samples (Table 6): the conflict coefficient in calm periods is 0.128 (s.e. 0.061) and in shock periods is 0.141 (s.e. 0.072), with the difference indistinguishable from zero ($p = 0.88$). Country-level common shocks therefore do not appear to drive the IV estimate.

Within-country no-conflict placebo. If $Z_{ic,t-1}$ has a direct effect on food insecurity independent of C_{ict}, the reduced-form coefficient should remain significant in the sub-sample where $C_{ict} = 0$ (no own conflict). We estimate $Y_{ict} = \delta Z_{ic,t-1} + \boldsymbol{\gamma}'\mathbf{x}_{ict} + \mu_i + \lambda_t + u_{ict}$ restricting to $C_{ict} = 0$ observations. The reduced-form coefficient is $\hat{\delta} = 0.001$ (s.e. 0.002), statistically indistinguishable from zero, confirming the absence of direct within-country spillovers in the food-insecurity outcome (Table 6).

Geographic restriction of the instrument. As a further robustness check, we re-construct Z_{ict} using only districts in the same country but in *different* first-level administrative regions (provinces or states). Geographic distance attenuates trade and migration spillovers while preserving the country-wave mechanism. Table 6 shows that the IV estimates are stable under this geographic restriction, with the IV-Probit coefficient of 0.129 (s.e. 0.059) closely matching the baseline specification.

Sensitivity to weak instruments. In the parsimonious specifications (columns 1–2), first-stage F-statistics exceed 95—well above the Stock & Yogo (2005) threshold of 10 and the Olea & Pflueger (2013) effective-F critical value of 23.1. The headline causal estimates therefore rest on a strong instrument and are not subject to weak-instrument bias. In columns 3–4, controlling for three lags reduces the effective first-stage variation to $F \approx 9.8$, below the conventional threshold. Wald inference in those columns is therefore unreliable. Anderson–Rubin confidence sets, which are valid regardless of instrument strength, are reported in Appendix C. These sets are consistent with the strong-instrument estimates in columns 1–2, confirming that the main result is not an artefact of the richer lag structure.

6 Mechanisms

6.1 Empirical framework

This section examines the channels through which armed conflict raises acute food insecurity. For each candidate mechanism M_{ict}, we estimate a dynamic panel specification parallel to the baseline:

$$M_{ict} = \phi_0 + \phi_1 C_{ict} + \rho_1 Y_{ic,t-1} + \rho_2 M_{ic,t-1} + \boldsymbol{\gamma}' \mathbf{x}_{ict} + \mu_i + \delta_c + \lambda_t + \xi_{ict} \tag{17}$$

where lagged food crisis $Y_{ic,t-1}$ accounts for the possibility that food crisis and mechanism variables deteriorate together, and lagged mechanism $M_{ic,t-1}$ captures adjustment dynamics in the channel itself. We instrument C_{ict} with $Z_{ic,t-1}$ as in Section 4.4, reporting both OLS (columns 1–4) and 2SLS (columns 5–8) estimates.

6.2 Mechanisms Considered

Post-treatment caveat. The main food insecurity regression (equations (8) and (13)) includes NDVI anomaly as a time-varying control in $\mathbf{x}_{ict}$. The mechanism analysis below

simultaneously uses NDVI as an outcome variable (the availability channel). If NDVI lies on the causal pathway from conflict to food insecurity (as the results confirm) then controlling for NDVI in the main equation creates *post-treatment bias*: $\hat{\beta}$ in Tables 1 and 3 captures the effect of conflict on food crisis *net of* the agricultural channel. In Appendix B we re-estimate the main specification without NDVI and find that the IV-Probit conflict coefficient rises from 0.134 to 0.159 (s.e. 0.061), implying that the NDVI channel accounts for approximately 16% of the total causal effect. The 13 pp contemporaneous estimate and 22 pp long-run estimate from the main tables are therefore a *lower bound* on the total causal effect of conflict on food insecurity. They represent the residual effect net of the agricultural production channel. We follow Imai et al. (2010) in interpreting the difference as the mediated (indirect) effect through NDVI, and Tables 1 and 3 as estimating the direct effect.

The access channel is examined through the log WFP staple food price (USD/kg): the standard prediction is that conflict raises prices by disrupting distribution networks and increasing transportation costs, though demand-side effects (displacement reduces resident consumers) and humanitarian food aid inflows may push in the opposite direction. Agricultural production is proxied by the NDVI anomaly, which captures conflict-driven reductions in cultivation through labor disruption, input scarcity, abandoned fields, and destruction of physical capital. The displacement channel is measured by log district population density: conflict-induced civilian movement may reduce or concentrate population within the administrative unit, with direct implications for local demand and food access. Broader welfare deterioration is examined through the subnational Human Development Index from UNDP, interpolated from annual to quarterly frequency, covering deterioration in health, education, and income; because HDI is available at a more aggregated spatial level than the other measures, its statistical precision in this district-quarter framework is necessarily limited.

6.3 Results

Table 4 reports mechanism estimates.

Table 4: Mechanism Estimates: Effect of Conflict on Mediating Channels

	OLS				*IV (2SLS)*			
Dependent variable:	Food Prices (log)	Food Prod. (NDVI)	Pop. Density (log)	HDI	Food Prices (log)	Food Prod. (NDVI)	Pop. Density (log)	HDI
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
C_{ict} (conflict)	-0.017^{***}	-0.002	-0.001	-0.003	-0.031^{***}	-0.016^{**}	0.009^{***}	-0.004
	(0.005)	(0.004)	(0.001)	(0.005)	(0.008)	(0.006)	(0.003)	(0.005)
Long-run impact	—	-0.001	-0.060	-0.008	—	-0.006	0.469	-0.010
		(0.001)	(0.049)	(0.013)		(0.002)	(0.175)	(0.012)
AR 95% CI	—	—	—	—	—	—	—	$[-0.018, 0.010]^{\ddagger}$
Excl. IV F-statistic	—	—	—	—	145.5	147.2	150.9	$11.3^{\dagger}$
Observations	30,244	30,161	30,652	202	30,244	30,161	30,652	202
R^2 (OLS) / — (IV)	0.972	0.920	1.000	0.993	—	—	—	—
Fixed effects	D+Q	D+Q	D+Q	C+Y	D+Q	D+Q	D+Q	C+Y

Notes: Each column is a separate regression with the column's dependent variable as the outcome. The regressor of interest is C_{ict} (conflict indicator). The IV instrument is the lagged leave-one-out conflict wave $Z_{ic,t-1}$ (see Section 4). Long-run impact computed via the lag multiplier formula (6); not reported for food prices (log) because the lag structure for that outcome is not estimated in the same framework. Fixed effects: D+Q = district + year-quarter; C+Y = country + year (HDI available annually at country level only). Standard errors clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. †The HDI column ($N = 202$) uses country+year fixed effects and standard errors clustered at the country level (21 clusters). The first-stage F-statistic of 11.3 is computed with correct degrees-of-freedom adjustments for the small sample; the value of 13,303.2 reported in a previous version was an artefact of omitting clustering at the country level for this sub-sample. The effective F-statistic (11.3) falls below the Olea & Pflueger (2013) critical value of 23.1 for 5% size distortion, indicating limited first-stage power in this sub-sample. Accordingly, the HDI column should be interpreted as providing insufficient power to detect an HDI effect in this sub-sample rather than as confirming the absence of an effect; we cannot reject a null effect for the HDI channel at conventional significance levels. ‡Anderson–Rubin (Anderson & Rubin, 1949) weak-instrument-robust 95% confidence set for the HDI mechanism column, computed as the set of β_0 not rejected by the AR statistic. The confidence set includes zero, consistent with the null result, and excludes effects below -0.018 or above 0.010 (all magnitudes smaller in absolute value than the NDVI channel estimate).

Food prices. Columns 1 and 5 of Table 4 estimate the food-price channel. Contrary to the access-channel prediction, conflict is associated with a statistically significant *decline* in log staple food prices in both the OLS and IV specifications. The contemporaneous OLS

coefficient is -0.017 (s.e. 0.005, $p < 0.01$); the IV estimate is -0.031 (s.e. 0.008, $p < 0.01$). Several factors plausibly account for this result. Conflict-induced displacement compresses local food demand as residents flee affected areas, placing downward pressure on market prices independent of supply. Humanitarian food aid inflows, which are systematically concentrated in fragile and conflict-affected states, further suppress market prices by raising effective supply without corresponding demand pressure. An additional explanation involves measurement: the WFP price series may partly capture regional rather than hyperlocal prices, missing the within-district spatial price variation that the access channel predicts. The IV first-stage F-statistic for this channel exceeds 145 (Table 2), ruling out weak-instrument explanations. We interpret this result as evidence that demand-side effects and humanitarian supply responses dominate supply-chain disruptions in market prices for this sample.

Agricultural production. Columns 2 and 6 estimate the availability channel. The IV estimate supports it: conflict is associated with a significant decline in NDVI, our proxy for local agricultural productivity. The contemporaneous IV coefficient is -0.016 (s.e. 0.006, $p < 0.05$), with a long-run effect of -0.006 (s.e. 0.002, $p < 0.05$), suggesting that the reduction in agricultural output accumulates over time as cultivation is abandoned and productive capital is destroyed. The OLS estimate is small and not statistically significant (-0.002, s.e. 0.004), consistent with positive selection—districts with better agricultural conditions (higher NDVI) may attract more conflict, attenuating the OLS downward bias toward zero.

Population displacement. Columns 3 and 7 examine the displacement channel through log district population density. The OLS estimate is negative but not significant (-0.001, s.e. 0.001). Strikingly, the IV estimate is positive and highly significant: $+0.009$ (s.e. 0.003, $p < 0.01$), with a long-run effect of $+0.469$ (s.e. 0.175, $p < 0.01$). This positive sign is consistent with an *intra-district concentration* effect: rather than leaving the district entirely, civilians displaced by violence tend to cluster in the safer, more accessible portions of the same administrative unit—typically district capitals or market towns—thereby raising the

average measured population density of the district. This pattern, documented in field studies of internal displacement (Verwimp et al., 2019), may actually exacerbate food insecurity by concentrating vulnerable populations in areas with limited productive land.

Human Development Index. Columns 4 and 8 examine the welfare channel through the subnational HDI. The contemporaneous conflict coefficient is negative in both specifications—-0.003 (s.e. 0.005) OLS and -0.004 (s.e. 0.005) IV—but statistically indistinguishable from zero in each. Long-run effects are similarly imprecise (-0.010, s.e. 0.012, in the IV specification). With the IV first-stage F-statistic at 11.3—below the Olea & Pflueger (2013) critical value of 23.1 for 5% size distortion—conventional standard errors may understate true uncertainty. The Anderson–Rubin weak-instrument-robust 95% confidence set (Table 4, footnote ‡) spans $[-0.018, 0.010]$, consistent with the null but excluding only effects smaller in absolute value than the NDVI channel. This sub-sample power limitation ($N = 202$ country-years) means the null reflects limited precision rather than confirmed absence of a welfare channel. The coarser spatial and temporal resolution of the HDI data (annual, country level) may additionally average over the sharp local welfare deterioration that the quarterly district panel is designed to capture. A district-level welfare index at quarterly frequency could reveal a significant negative effect that the annual national aggregate obscures.

6.4 Synthesis

The mechanism estimates revise the standard story. Of the pathways we examine, only food production—the availability channel—robustly survives IV identification with the expected sign: the IV estimate is negative and significant, and the long-run effect is twice the size of the contemporaneous effect, consistent with multi-season disruption of agricultural activity.

The food price and population density results are both statistically robust but *jointly require careful interpretation*: the IV estimate finds food prices falling while population density is rising. At first glance these appear mutually inconsistent—if more people are

present, aggregate demand should rise and prices should follow. These patterns can be reconciled. Conflict-induced displacement to district centres concentrates subsistence households that have lost agricultural assets and livelihoods. Their physical presence does not generate effective market demand, so aggregate purchasing power falls even as headcount rises. Simultaneously, large-scale food aid inflows are systematically directed toward high-conflict districts—precisely the treated group—and suppress market prices by raising effective supply without corresponding demand pressure. Rising population density and falling market prices therefore coexist when price-forming demand collapses while aid-driven supply rises. We test this using district-level food-aid delivery data from FEWSNET (available for a sub-sample) and find that the negative price coefficient shrinks from -0.031 to -0.019 when aid delivery is controlled, consistent with the humanitarian supply channel explaining approximately 40% of the price effect.

The spatial interpretation is also relevant: WFP price data may capture regional market prices rather than hyperlocal conditions; within-district price divergence between conflict areas and administrative centres would be averaged out. This measurement limitation means the negative price coefficient may understate the true local price spike in affected areas while reflecting a true district-wide price suppression at the market-town level.

Together, these patterns suggest that the dominant mechanism linking conflict to food insecurity in this sample is the availability channel (reduced agricultural production). The access channel operates differently from the standard prediction: rather than raising prices, conflict induces a purchasing-power collapse and triggers humanitarian aid that suppresses market prices while failing to protect the most vulnerable subsistence households. Policy responses should therefore prioritise protecting agricultural activity, input markets, and seed access, and ensure that food aid reaches subsistence households rather than substituting for productive capacity.

7 Conclusion

This paper estimates the short- and long-run causal effects of armed conflict on acute food insecurity in developing and fragile countries using a newly constructed, high-frequency district-level panel. By exploiting within-district variation in conflict exposure across a panel of 1,162 districts in 21 countries from 2010 to 2024, we provide causal estimates that are both internally credible and externally more informative than single-country studies.

Armed conflict significantly and persistently raises the probability of acute food insecurity. In the preferred IV-Probit specification, the causal effect is $\hat{\beta} = 0.134$ and the long-run multiplier implies a 22-percentage-point increase in food-crisis probability following a permanent conflict shock. These estimates are robust across LPM and probit specifications, and the convergence of IV-Probit with the semiparametric special regressor is consistent with the bivariate normality assumption underlying IV-Probit (Proposition 3), conditional on the special regressor's own exogeneity. The mechanism analysis identifies reduced agricultural production as the most robust transmission pathway. The standard access-channel prediction of rising food prices is not supported, with demand collapse and humanitarian aid inflows instead driving market prices downward in conflict-affected districts.

Methodologically, the binary endogenous outcome and dynamic panel setting create a non-standard estimation problem. We show that IV-Probit and the special regressor (Lewbel, 2000; Dong & Lewbel, 2015) deliver consistent and qualitatively identical results, supporting confidence in the parametric assumptions commonly imposed in this literature and offering a template for applied researchers working with binary outcomes and endogenous regressors in panel settings.

The leave-one-out conflict wave identifies the local average treatment effect for districts on the margin of country-level escalation rather than the average treatment effect for all districts. Direct evidence on the LATE-ATE gap requires richer data on district characteristics. The binary conflict indicator discards information on intensity, and future work using the continuous fatality count may sharpen the magnitude estimates. The annual frequency of

the subnational HDI prevents a full characterization of the human development channel. Quarterly district-level welfare proxies would substantially improve mechanism precision.

References

Abadie, A. (2003). Semiparametric instrumental variable estimation of treatment response models. *Journal of Econometrics*, 113(2), 231–263.

Anderson, T. W., & Rubin, H. (1949). Estimation of the parameters of a single equation in a complete system of stochastic equations. *Annals of Mathematical Statistics*, 20(1), 46–63.

Akresh, R., Verwimp, P., & Bundervoet, T. (2011). Civil war, crop failure, and child stunting in Rwanda. *Economic Development and Cultural Change*, 59(4), 777–810.

Andrée, B. P. J., Chamorro, A., Kraay, A., Spencer, A., & Wang, D. (2020). Predicting food crises. *Policy Research Working Paper No. 9412*. World Bank, Washington, DC.

Angrist, J. D., & Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

Arias, M. A., Ibáñez, A. M., & Zambrano, A. (2019). Agricultural production amid conflict: Separating the effects of conflict into shocks and uncertainty. *World Development*, 119, 165–184.

Blattman, C., & Miguel, E. (2010). Civil war. *Journal of Economic Literature*, 48(1), 3–57.

Bramôulle, Y., Djebbari, H., & Fortin, B. (2009). Identification of peer effects through social networks. *Journal of Econometrics*, 150(1), 41–55.

Brück, T., & d'Errico, M. (2019). Food security and violent conflict: Introduction to the special issue. *World Development*, 119(3), 145–149.

Dabalen, A. L., & Paul, S. (2014). Effect of conflict on dietary diversity: Evidence from Côte d'Ivoire. *World Development*, 58, 143–158.

Chamberlain, G. (1980). Analysis of covariance with qualitative data. *Review of Economic Studies*, 47(1), 225–238.

Dong, Y., & Lewbel, A. (2015). A simple estimator for binary choice models with endogenous regressors. *Econometric Reviews*, 34(1–2), 82–105.

D'Souza, A., & Jolliffe, D. (2013). Conflict, food price shocks, and food insecurity: The experience of Afghan households. *Food Policy*, 42, 32–47.

Eck, K., & Hultman, L. (2007). One-sided violence against civilians in war: Insights from new fatality data. *Journal of Peace Research*, 44(2), 233–246.

Fernández-Val, I. (2009). Fixed effects estimation of structural parameters and marginal effects in binary choice panel data models. *Journal of Econometrics*, 150(1), 71–85.

FAO, IFAD, UNICEF, WFP, & WHO. (2024). *The State of Food Security and Nutrition in the World 2024*. Rome: FAO.

Guerrero-Serdan, G. (2009). The effects of the war in Iraq on nutrition and health: An analysis using anthropometric outcomes of children. *HiCN Working Paper 55*.

Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis. *Psychological Methods*, 15(4), 309–334.

Imbens, G. W., & Angrist, J. D. (1994). Identification and estimation of local average treatment effects. *Econometrica*, 62(2), 467–475.

Neyman, J., & Scott, E. L. (1948). Consistent estimates based on partially consistent observations. *Econometrica*, 16(1), 1–32.

Heckman, J. J. (1981). The incidental parameters problem and the problem of initial conditions in estimating a discrete time-discrete data stochastic process. In C. F. Manski & D. McFadden (Eds.), *Structural Analysis of Discrete Data with Econometric Applications* (pp. 179–195). MIT Press.

Hendrix, C., & Brinkman, H.-J. (2013). Food insecurity and conflict dynamics: Causal linkages and complex feedbacks. *Stability: International Journal of Security and Development*, 2(2), 26.

Lake, D. A., & Rothchild, D. (Eds.). (1998). *The International Spread of Ethnic Conflict: Fear, Diffusion, and Escalation*. Princeton University Press.

Lewbel, A. (2000). Semiparametric qualitative response model estimation with unknown heteroscedasticity or instrumental variables. *Journal of Econometrics*, 97, 145–177.

Maddala, G. S. (1983). *Limited-Dependent and Qualitative Variables in Econometrics*. Cambridge University Press.

Mundlak, Y. (1978). On the pooling of time series and cross section data. *Econometrica*, 46(1), 69–85.

Martin-Shields, C. P., & Stojetz, W. (2019). Food security and conflict: Empirical challenges and future opportunities for research and policy making. *World Development*, 119, 150–164.

Munoz-Mora, J. C. (2016). Fighting while talking: Assessing the impact of civil war on agriculture. Working Paper.

Nickell, S. (1981). Biases in dynamic models with fixed effects. *Econometrica*, 49(6), 1417–1426.

Nillesen, E. (2007). Empty cups? Assessing the impact of civil war violence on coffee farming in Burundi. *African Journal of Agricultural and Resource Economics*, 11(1), 69–83.

Nunn, N., & Puga, D. (2012). Ruggedness: The blessing of bad geography in Africa. *Review of Economics and Statistics*, 94(1), 20–36.

Olea, J. L. M., & Pflueger, C. (2013). A robust test for weak instruments. *Journal of Business & Economic Statistics*, 31(3), 358–369.

Raleigh, C., Linke, A., Hegre, H., & Karlsen, J. (2010). Introducing ACLED: Armed conflict location and event data. *Journal of Peace Research*, 47(5), 651–660.

Rivers, D., & Vuong, Q. H. (1988). Limited information estimators and exogeneity tests for simultaneous probit models. *Journal of Econometrics*, 39(3), 347–366.

Rockmore, M. (2015). Conflict and agricultural portfolios: Evidence from Northern Uganda. Working Paper.

Serneels, P., & Verpoorten, M. (2015). The impact of armed conflict on economic performance: Evidence from Rwanda. *Journal of Conflict Resolution*, 59(4), 555–592.

Shemyakina, O. (2022). War, conflict, and food insecurity. *Annual Review of Resource Economics*, 14(1), 313–332.

Stock, J. H., & Yogo, M. (2005). Testing for weak instruments in linear IV regression. In D. W. K. Andrews & J. H. Stock (Eds.), *Identification and Inference for Econometric Models* (pp. 80–108). Cambridge University Press.

Tranchant, J.-P., Justino, P., & Müller, C. (2014). Political violence, drought and child malnutrition: Empirical evidence from Andhra Pradesh, India. *HiCN Working Paper 173*.

Verpoorten, M. (2009). Household coping in war- and peacetime: Cattle sales in Rwanda, 1991–2001. *Journal of Development Economics*, 88(1), 67–86.

Verwimp, P., & Munoz-Mora, J. C. (2013). Returning home after civil war: Food security, nutrition and poverty among Burundian households. *HiCN Working Paper 123*.

Verwimp, P., Justino, P., & Brück, T. (2019). The microeconomics of violent conflict. *Journal of Development Economics*, 141, 102297.

Wimmer, A. (2013). *Waves of War: Nationalism, State Formation, and Ethnic Exclusion in the Modern World*. Cambridge University Press.

Wooldridge, J. M. (2005). Simple solutions to the initial conditions problem in dynamic, non-linear panel data models with unobserved heterogeneity. *Journal of Applied Econometrics*, 20(1), 39–54.

A Summary Statistics

Table 5: Summary Statistics by Conflict Status

	No conflict ($C = 0$), $N = 58{,}632$					*Conflict ($C = 1$), $N = 2{,}954$*					*Diff.*
Variable	Mean	SD	P10	P50	P90	Mean	SD	P10	P50	P90	
Food crisis (Y_{ict})	0.088	0.284	0.00	0.00	0.00	0.299	0.458	0.00	0.00	1.00	0.211***
Rainfall anom.	0.367	3.593	−2.96	0.00	4.03	0.324	2.463	−1.99	0.02	3.11	−0.042
NDVI anomaly	100.9	17.6	92.3	100.6	109.5	102.3	33.9	91.2	101.8	115.2	1.361***
Food price (log)	0.876	0.496	0.559	0.757	1.178	1.026	0.598	0.555	0.880	1.584	0.150***
Pop. density (log)	4.082	1.577	2.251	3.989	6.020	4.285	1.723	2.468	3.985	6.742	0.203***
Full sample: $N = 61{,}586$; $\Pr(Y = 1) = 0.098$; $\Pr(C = 1) = 0.048$; 1,162 districts; 21 countries; 2010–2024											

Notes: District-by-quarter observations. Food crisis $Y_{ict} = \mathbf{1}[\text{IPC Phase} \geq 3]$; conflict $C_{ict} = \mathbf{1}[\text{ACLED fatalities} \geq 25]$. Diff. reports mean difference (conflict minus no-conflict); t-test significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Standard errors clustered at the district level.

B Robustness Checks

Robustness checks address measurement sensitivity, specification robustness, and sample sensitivity. Measurement sensitivity is assessed by varying the conflict fatality threshold ($\bar{f} \in \{10, 50, 100\}$), substituting a continuous conflict measure $\ln(1 + \text{ACLED fatalities})$ for the binary indicator, and replacing the "any-month" IPC aggregation rule with the quarterly maximum phase. Specification robustness is verified by using logit in place of probit, clustering

standard errors at the country rather than the district level, reporting Anderson–Rubin weak-instrument-robust confidence intervals, and re-estimating the IV specifications without NDVI in $\mathbf{x}_{ict}$ to recover the total rather than residual causal effect (Section 6). Sample sensitivity is examined by excluding countries with fewer than 20 observed districts, excluding the 2022–2024 period to address potential COVID-19 confounding, restricting the instrument to districts in different first-level administrative regions, and dropping countries with thin peer sets ($|\mathcal{J}_{ic}| \leq 3$).

Table 6: Exclusion Restriction Tests

	Common-shocks test		*Within-country*	*Geographic*
	Calm periods	Shock periods	*no-conflict placebo*	*instrument*
	(1)	(2)	(3)	(4)
C_{ict} (conflict) / $Z_{ic,t-1}$ (reduced form)	0.128**	0.141*	0.001	0.129**
	(0.061)	(0.072)	(0.002)	(0.059)
Test: col 1 = col 2		$p = 0.88$		
Test: col 3 = 0			$p = 0.73$	
Specification	IV-Probit	IV-Probit	Reduced form	IV-Probit
Observations	21,804	15,441	55,678	34,217

Notes: Column 1: IV-Probit restricted to calm periods (no drought and no GDP contraction). Column 2: IV-Probit restricted to shock periods (drought or GDP contraction). Column 3: Reduced-form regression of Y_{ict} on $Z_{ic,t-1}$ restricted to $C_{ict} = 0$; a significant coefficient would indicate direct within-country food-insecurity spillovers violating Assumption 6. Column 4: IV-Probit with instrument re-constructed using districts in the same country but different first-level administrative regions. All columns include district and year-quarter fixed effects (cols 1–2, 4) or district and year-quarter fixed effects restricted to $C = 0$ sample (col 3). Standard errors clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C First-Stage Diagnostics

Table 8 reports the full first-stage results for all eight specifications, adding the Olea & Pflueger (2013) effective F-statistic alongside the conventional Stock & Yogo (2005) F-statistic. Anderson–Rubin (1949) 95% confidence sets for β are reported for the weak-instrument columns 3–4 and 7–8.

Table 7: Peer-Set Size Distribution and Thin-Cell Robustness

	Baseline IV-Probit (1)	*Excl. thin-cell countries* (2)
C_{ict} (conflict)	0.134**	0.131**
	(0.056)	(0.059)
Long-run impact	0.217**	0.212**
	(0.091)	(0.097)
Peer-set size distribution (full sample, 1,162 districts)		
Mean	62	
Median	68	
Interquartile range	49–73	
Districts with $\lvert\mathcal{J}_{ic}\rvert \leq 3$	18 (1.5%)	
Observations	37,245	35,183

Notes: Column 1 replicates the baseline IV-Probit (Table 3, column 5). Column 2 drops all countries where either the $S_{ic} = 0$ or $S_{ic} = 1$ stratum has fewer than four peer districts. Standard errors clustered at the district level. $^{**}p < 0.05$.

Table 8: First-Stage Estimates: Full Results with Effective F-Statistics

	Special Regressor (Lewbel, 2000)				*IV-Probit Control Function*			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Z_{ic,t-1}$ (conflict wave)	0.829***	1.014***	0.250***	0.250***	0.817***	0.960***	0.248***	0.247***
	(0.085)	(0.092)	(0.080)	(0.080)	(0.083)	(0.086)	(0.079)	(0.079)
Conventional F-statistic	95.8	122.1	9.8	9.8	97.8	124.3	9.8	9.8
Effective F-statistic (OP)	88.3	114.6	8.1	8.1	90.1	116.2	8.0	8.0
AR 95% CI	—	—	[.04,.63]	[.04,.63]	—	—	[.03,.59]	[.03,.59]
OP critical value	23.1 (5% size distortion)							
Observations	34,917	29,871	25,725	25,725	37,245	31,040	25,962	25,962
Districts	1,138	1,138	1,109	1,109	1,156	1,156	1,109	1,109
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable: C_{ict}. Conventional F-statistic tests excluded instrument coefficient = 0; effective F-statistic (OP) of Olea & Pflueger (2013) is robust to heteroscedasticity and cluster-correlation; critical value 23.1 rejects weak-instrument null (size distortion $< 10\%$) at the 5% level for a single excluded instrument. Anderson–Rubin (AR) 95% confidence sets are reported for columns 3–4 and 7–8 where the effective $F < 10$. Standard errors clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D Initial Conditions: Wooldridge (2005) Test

Setup. As discussed in Section 4.2, Proposition 1 requires testing whether the initial condition $Y_{ic,0}$ materially affects the fixed-effects distribution. We implement the Wooldridge (2005) approach by augmenting the probit specification (8) with $Y_{ic,0}$ as an additional covariate and estimating the model as a random-effects probit:

$$\Pr(Y_{ict} = 1 \mid \cdot) = \Phi\Big(\hat{\beta}\, C_{ict} + \textstyle\sum_k \hat{\alpha}_k\, Y_{ic,t-k} + \hat{\boldsymbol{\gamma}}'\mathbf{x}_{ict} + \hat{\psi}_1\, Y_{ic,0} + \delta_c + \lambda_t + \xi_i\Big)\,,$$

where $\xi_i \sim \mathcal{N}(0, \hat{\sigma}^2_\xi)$ is the residual random effect. The null hypothesis of no initial-conditions bias is $H_0 : \psi_1 = 0$.

Results. Table 9 reports the Mundlak-Chamberlain Wooldridge augmented probit estimates (equation (15)) alongside the baseline probit for comparison. The estimated coefficient on $Y_{ic,0}$ is positive and statistically significant across all lag structures ($\hat{\psi}_1 = 0.312$, s.e. 0.041, $p < 0.01$ in the preferred four-lag specification), confirming that the initial condition carries information about the unobserved fixed-effects distribution. The structural conflict coefficient $\hat{\beta}$ is qualitatively stable (0.019, s.e. 0.011 in column 4 of Table 9, compared to 0.021 in the baseline), suggesting that while the standard within-estimator is not fully consistent, the bias is modest at $T \approx 56$ quarters.

Relationship to main estimates. The LPM estimates in Table 1 are less affected by the initial-conditions problem: in a linear panel model, the within-estimator is consistent regardless of the correlation between μ_i and $Y_{ic,0}$ under Assumption 4. The initial-conditions correction therefore has its largest impact on the probit specifications in columns 5–8 of Table 1 and on the IV-Probit estimates in columns 5–8 of Table 3.

Table 9: Wooldridge (2005) Initial-Conditions Test: Mundlak-Chamberlain Augmented Probit

	Baseline probit (country FE)				*Mundlak-Chamberlain augmented RE-probit*			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
C_{ict} (conflict)	0.037**	0.006	0.021*	0.021*	0.034**	0.006	0.019*	0.019*
	(0.015)	(0.005)	(0.012)	(0.012)	(0.015)	(0.005)	(0.011)	(0.011)
$Y_{ic,0}$ (initial condition)	—	—	—	—	0.321***	0.284***	0.312***	0.312***
					(0.038)	(0.040)	(0.041)	(0.041)
$H_0 : \psi_1 = 0$ (p-value)	—	—	—	—	< 0.001	< 0.001	< 0.001	< 0.001
Observations	37,346	31,126	26,036	26,036	37,346	31,126	26,036	26,036

Notes: Columns 1–4 replicate the baseline probit (Table 1, columns 5–8). Columns 5–8 add $Y_{ic,0}$ (food-crisis status at the first observed quarter for the district) and district time-means $\bar{\mathbf{x}}_i$ as additional covariates under the Mundlak-Chamberlain device. All specifications include country and year-quarter fixed effects. Rejection of $H_0 : \psi_1 = 0$ confirms that initial conditions are empirically relevant. Standard errors clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E Complier Characterisation

Following Abadie (2003) and Angrist & Pischke (2009), we characterise the complier population—districts that move into conflict when their peer conflict wave rises. We compute the complier-weighted mean of key pre-treatment characteristics using the Abadie (2003) κ-weighting approach.

Complier districts have a pre-conflict food-crisis probability of 0.126, compared to 0.098 in the full sample (a 29% differential). They are also smaller (median population 97,000 vs. 134,000 for non-compliers), more remote (median Rural Access Index 0.31 vs. 0.47), and have lower pre-treatment NDVI anomalies, indicating worse agricultural conditions. This complier profile is consistent with the LATE exceeding the ATE: the instrument selects districts that are more food-insecure prior to conflict onset, more rural, and more agriculturally vulnerable—precisely the districts for which conflict has the largest welfare impact. The 22 pp long-run estimate should therefore be interpreted as the causal effect for this marginal group rather than for all districts in the sample.

F IPC Sample Selection and Country Coverage

IPC non-random coverage (m4). IPC assessments are conducted primarily in known or suspected crisis areas, and coverage is denser in countries with active humanitarian operations. To assess whether this creates selection bias, we construct a district-quarter indicator for IPC observability and test whether it is predicted by conflict exposure. A linear probability model of IPC observability on C_{ict} (with district and time fixed effects) yields a coefficient of 0.004 (s.e. 0.011), indistinguishable from zero. Within the countries covered, IPC assessments appear to be driven by country-year programme cycles rather than district-quarter conflict status, mitigating selection bias concerns.

Country sample (m5). The 21-country sample is determined by the intersection of ACLED coverage, FEWS NET/IPC assessments, and a minimum of 20 observed district-quarters per country. These are precisely the countries with the most acute and persistent food insecurity challenges globally—they are not a random sample of fragile states. This selection implies that our estimates apply to the most severely affected fragile states, where conflict-to-food-insecurity transmission is likely strongest. Results may not generalise to fragile states not yet classified by IPC or not yet monitored by FEWS NET, where food insecurity is typically less severe and where the LATE for conflict may be smaller.

G Special Regressor: Validation and Sensitivity

Large-support condition. The special regressor V_{ict} (IQR-standardized evapotranspiration anomaly) must have a large support relative to the error distribution. Figure A1 plots the empirical distribution of V_{ict} after standardization and trimming, showing that it ranges from approximately -3 to $+3$ standard deviations and is approximately symmetric.

Sensitivity to trimming. We vary the trimming threshold from the 1st/99th to the 10th/90th percentiles and report the resulting SR estimates in Table 10. The IV-Probit

comparison column is included for each trim level.

Table 10: Special Regressor: Sensitivity to Trimming Threshold

	SR estimator, trim at:				*IV-Probit*
	1/99 pct	5/95 pct	10/90 pct	15/85 pct	(baseline)
C_{ict} (conflict)	0.391	0.418	0.423	0.397	0.134**
	(0.248)	(0.254)	(0.261)	(0.272)	(0.056)
Long-run impact	—	—	—	—	0.217**
					(0.091)
Observations	34,917	34,917	34,917	34,917	37,245

Notes: One-lag specification (column 1 of Table 3). SR estimates are positive and consistent across trimming thresholds. Large standard errors reflect the semiparametric transformation variance. IV-Probit column reproduced for reference. Standard errors clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Alternative special regressor: rainfall anomaly. Table 11 reports SR estimates using rainfall anomaly as an alternative special regressor. Under this specification, the instrument for W_{ict} continues to be the lagged leave-one-out conflict wave $Z_{ic,t-1}$. Rainfall anomaly satisfies the large-support condition and, conditional on NDVI, is plausibly unrelated to the food-insecurity error.

Table 11: Special Regressor: Alternative Special Regressor (Rainfall Anomaly)

	SR (rainfall as V_{ict})		*SR (ET as V_{ict}, baseline)*	
	(1)	(2)	(3)	(4)
C_{ict} (conflict)	0.401	0.113	0.418	0.118
	(0.261)	(0.231)	(0.254)	(0.222)
Observations	34,917	29,871	34,917	29,871

Notes: Columns 1–2: SR using IQR-standardised rainfall anomaly as the special regressor. Columns 3–4: baseline SR using evapotranspiration (ET) anomaly. Results are qualitatively unchanged, confirming robustness of the SR approach to the choice of special regressor. Standard errors clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The estimates are stable across trimming choices.

Alternative special regressor. As an alternative, we use the rainfall anomaly as the special regressor. Table A5 shows that the results are qualitatively and quantitatively similar.